

Things You Wanted to Know!

Q1. I want to start making electronic items for which I need soldering machines, stations and other components like LEDs and resistors. I request you to send me some details regarding the same.

Jaswinder Kaur
Ropar, Punjab

A1. You will find the necessary information on sources of supply of components and instruments in EFY Electronics Industry Directory. You can also get information online from our website www.ele2b.com

For soldering machines, you need to know parameters like tip temperature, tip-to-ground resistance, solder pot temperature and ambient temperature, among others.

Parameters for selection of LEDs are colour, viewing-angle characteristics, forward and reverse voltage luminous intensity and package, among others.

Resistor parameters to be considered include type, wattage, tolerance, voltage rating, temperature coefficient, stability and others.

Q2. What should be the capacity of the battery for a 100-watt solar panel with three-hour run time?

Ram Mohanty
Cuttack, Odisha

A2. For 100W load and three-hour runtime with a margin of 25 per cent, the rating of the inverter would be 125W. Capacity of the battery should be $3 \times 125 = 375\text{Wh}$. So we have to use two batteries of 200Wh ($2 \times 200 = 400\text{Wh} > 375\text{Wh}$) each.

Q3. I would like to know the features of 3-bit-per-cell (X3) technology.

Ashish Mulajkar
Pune, Maharashtra

A3. 3-bit-per-cell (X3) technology relates to flash memory structure. Flash memory has been available for the last few years with single-level cells having 1-bit storage per cell (SLC) and 2-bit per cell multi-level cell (MLC).

Multi-level cell flash is a type of flash memory that stores more than one bit per cell. The ever-increasing digital data places an increased demand for NAND flash memory, which has led to the development of X3 technology. It is co-developed by SanDisk and Toshiba, which shares with SanDisk the development and manufacturing of advanced flash memories.

The memory-tiered structure is shown in the figure below. X3 technology requires no change in the basic design of flash memory from a manufacturing viewpoint. With the same process technology, X3 provides more bits per wafer and lower cost per bit. Thus, the cost reductions per

sumption is less in SLC/MLC. MLC flash is less expensive than SLC, which makes it a popular choice for consumer-grade solid-state storage.

Q4. The 3-pole electronic relay with sufficient current rating is available, but generally it is not used to operate the 3-phase, 400V AC, 5HP induction motor. Electric contactors are widely used to operate it. Please explain why relays are not used?

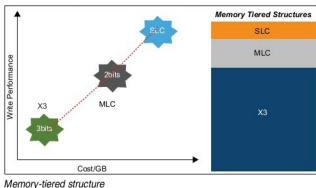
S. Chinnadurai
Salem-DT, Tamil Nadu

A4. Relays and contactors are electromagnetic switches. The only difference is that relays are usually used for low-voltage applications, while contactors are used for higher-voltage applications. Contactors with accessories like start, stop buttons and overload relays provide integrated solutions for starting and controlling operations of 3-phase AC motors.

Direct online starters using contactors are available as single units, which can easily be mounted on control panels. Contactors have larger-size contact strips, which are capable of carrying more current. The distance between

contact strips is also more, therefore capable of withstanding higher voltages.

Relays are normally used in appliances below 5kW, while contactors are preferred when the appliance is heavier in terms of current requirement. Further, relays are cumbersome to use when compared with contactors.



bit can be significant using X3. This technology also maintains low cost as shown in the figure.

Write performance is better in SLC/MLC as shown in the figure. Read performance in X3 does not differ much between SLC and MLC. The more bits the cell has, the fewer write cycles it will have. For example, a 2-bit MLC cell is good for about 3000 to 10,000 write operations before it begins to fail, while a 3-bit MLC cell would only have 300 to 3000 write cycles. Once a cell is written to its limit, it starts to forget what is stored and can corrupt the data. Power con-

Answers compiled by EFY joint director (training), Col. N.C. Pande (Retd). Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender